

Introduction

The process of automatically determining the natural language (e.g., English, Spanish, French) of a given piece of text or spoken content. For humans, language detection is a skill that can be developed through frequent exposure to a particular language and by understanding its patterns and sound composition. In this project, we will teach a machine to develop spoken-language detection using live audio or uploaded audio files.

Objective

The primary objective of this project is to develop and implement a machine learning model that can accurately identify a language spoken in an audio input.

This will involve:

1. **Data Acquisition and Preprocessing:** Curating a diverse dataset of spoken audio clips across multiple target languages, and performing necessary preprocessing steps (e.g., noise reduction, feature extraction like Mel-Frequency Cepstral Coefficients (MFCCs)).
2. **Model Selection and Training:** Evaluating and implementing appropriate deep learning architectures (e.g., Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs)/LSTMs) optimized for sequential audio data.
3. **Real-Time and File-Based Detection:** Building an interface or system that can perform language identification on both pre-recorded audio files (live audio streams if time allows).
4. **Performance Evaluation:** Achieving a high degree of accuracy (e.g., over 90%) in distinguishing between the target languages and providing a quantitative analysis of the model's performance metrics (precision, recall, F1-score).

Timeline

The timeline provided here shows a general overview of the timeline, but might not match the exact development process.

Task	Start Date	End Date	Duration (Days)	Notes
Data Selection	April 01	April 05	5	Curating and selecting the target audio datasets.

Understanding Audio Data Processing	April 05	April 10	6	Feature extraction (MFCCs), preprocessing, and noise reduction.
Model Training	April 10	April 20	11	Implementing and training deep learning models (CNN/RNN/LSTMs)
Model Evaluation	April 20	April 25	6	Testing, hyperparameter tuning, and performance metric calculation.
Work Finalization & Presentation Prep	April 25	May 07	13	System integration, documentation, and creating presentation slides.

Reference

[Kaggle: Spoken Language Identification](#)

[Mozilla data collective: common voice](#)

[Pytorch documentation](#)